🔹 1. Using Aggregate Functions (SUM, COUNT, AVG)

Aggregate functions perform calculations on multiple rows and return a single value.

- **SUM(column_name):** Adds all values in a numeric column.
- **COUNT(column_name |):* Counts number of rows or non-null values.
- **AVG(column_name):** Calculates average of numeric column values.

Example:

sql

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SELECT SUM(salary) AS total_salary FROM employees; SELECT COUNT(*) FROM employees WHERE department = 'Sales'; SELECT AVG(age) FROM students;

📌 2. GROUP BY and HAVING Clauses for Grouped Data

- **GROUP BY:** Groups rows that have the same values in specified columns.
- Often used with aggregate functions to compute values per group.

Syntax:

sql

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```
SELECT column1, AGG_FUNC(column2) FROM table GROUP BY column1;
```

Example:

sql

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SELECT department, COUNT(*) AS employee_count FROM employees GROUP BY department;

- **HAVING:** Filters groups after aggregation (like WHERE but for groups).

Example:

sql

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SELECT department, COUNT(*) AS employee_count FROM employees GROUP BY department HAVING COUNT(*) > 10;

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3. Subqueries and Nested Queries

- A **subquery** is a query nested inside another query.
- Used in SELECT, FROM, WHERE, or HAVING clauses.
- Helps break complex problems into simpler parts.

Example in WHERE clause:

sql

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```
SELECT name, salary FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);
```

Example in FROM clause:

sql

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```
SELECT department, avg_salary FROM (SELECT department, AVG(salary) AS avg_salary FROM employees GROUP BY department) AS dept_avg
WHERE avg_salary > 50000;
```
